SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Q1. Compare Star Topology and Bus Topology
Answer:
Star topology and bus topology differ mainly in **cost, reliability, performance, and cabling**.

Factor	Star Topology	Bus Topology
Cost	Higher because a central switch/hub is required	Lower because one backbone cable is used
Cabling	More cables are required	Less cabling
Reliability	High; failure of one node does not affect others	Low; backbone failure can affect the entire network
Performance	Better because communication is controlled by the central switch	Decreases as more devices share the backbone
Troubleshooting	Easy to identify individual cable/device failures	Difficult to locate backbone faults
Scalability	Easier to expand by adding switch ports	Limited due to shared backbone

Star topology is therefore more suitable for **modern offices and organizations** where reliability, performance and easy management are important. Bus topology is more suitable for **small, low-cost networks** where simplicity is the main requirement. The source specifically notes that star has better reliability and performance, while bus has lower installation cost.
Conclusion: Star topology provides better overall performance and reliability, whereas bus topology is advantageous mainly because of its low cost and simple structure.

Q2. Compare Mesh Topology and Star Topology in Terms of Fault Tolerance
Answer:
Mesh topology provides **higher fault tolerance** than star topology because each node has multiple possible communication paths.
In a **mesh topology**, if one link fails, data can travel through another available path. Therefore, the failure of a single connection does not necessarily interrupt communication.
In a **star topology**, failure of an individual cable or end device affects only that device. However, the central switch or hub is a critical point. If the central device fails, communication between all connected devices can be interrupted.

Factor	Mesh	Star
Fault tolerance	Very high	Moderate
Alternate paths	Multiple	Generally unavailable
Single point of failure	Very low	Central switch/hub
Reliability	Very high	High
Complexity	High	Low
Suitable for	Critical networks	Offices and general networks

Therefore, **mesh topology is more appropriate for critical environments** requiring high availability, while star topology provides a simpler and more economical solution.
Conclusion: Mesh is superior in fault tolerance because alternate paths allow communication to continue even when a link fails.

Q3. Compare Bus and Mesh Topologies in Terms of Installation Complexity
Answer:
Bus topology is much simpler to install than mesh topology.
A **bus topology** uses a single backbone cable to which multiple computers are connected. It requires fewer cables and connections, making installation simple and inexpensive.
A **mesh topology** requires each node to have connections with multiple or all other nodes. As the number of devices increases, the number of connections increases significantly. This makes installation more complex, time-consuming and expensive.

Factor	**Bus**	**Mesh**
Installation	Simple	Complex
Cabling	Minimal	Very high
Number of connections	Low	Very high
Cost	Low	High
Expansion	Relatively difficult	Complex
Maintenance	Simple initially	More difficult

For example, a small classroom with a few computers can use a simple bus arrangement when cost is important, whereas a critical organization may choose mesh despite its complexity because of its reliability.
Conclusion: Bus topology is easier, cheaper and faster to install, while mesh topology provides greater reliability at the cost of increased installation complexity.

Q4. Compare Hybrid and Star Topologies in Terms of Scalability
Answer:
Hybrid topology is more scalable than star topology because it combines two or more topology types according to the requirements of different network sections.
In a star topology, all devices are connected to a central switch or hub. Expansion depends on the number of available ports. Once the switch reaches its capacity, additional networking hardware is required.
Hybrid topology allows organizations to create different network sections using star, mesh, bus or other topologies. New segments can be added without significantly modifying existing sections.

Factor	**Hybrid**	**Star**
Scalability	Very high	Moderate to high
Expansion	Flexible	Depends on switch capacity
Network structure	Multiple topologies	Single topology
Flexibility	Very high	Moderate
Large networks	Highly suitable	Requires additional switches
Cost	Depends on design	Relatively predictable

For example, a **university campus** can use a hybrid structure with a redundant backbone and star topology within individual buildings. This allows new buildings or departments to be added without disturbing existing network segments.
Conclusion: Hybrid topology is better for large and growing organizations because it provides greater flexibility and scalability than a single star topology.

Q5. Compare Star, Mesh, Bus and Hybrid Topologies in Terms of Maintenance
Answer:
Maintenance requirements vary significantly among star, mesh, bus and hybrid topologies.

Topology	Maintenance	Troubleshooting	Main Advantage	Main Limitation
Star	Easy	Easy	Faults can be isolated to individual nodes	Central switch is critical
Mesh	Difficult	Complex	Very high reliability	Many connections to manage
Bus	Difficult	Difficult	Simple and inexpensive	Backbone failure affects network
Hybrid	Moderate	Requires skilled administration	Flexible and scalable	Multiple topology types increase complexity

Star topology is easiest to maintain because a faulty node or cable can usually be identified and isolated without affecting the rest of the network.
Mesh topology is highly reliable but difficult to maintain because it contains a large number of connections.
Bus topology can be difficult to troubleshoot because a fault in the shared backbone can affect many or all devices.
Hybrid topology has moderate maintenance complexity. Its flexibility is useful, but administrators must understand and monitor the different topology sections.
Conclusion: For easy maintenance, **star topology is the best choice**. Mesh provides the highest reliability but requires more maintenance effort, bus is inexpensive but difficult to troubleshoot, and hybrid provides flexibility with moderate maintenance complexity.